

Padhmavathy Cebolu Srinivasan

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EDUCATION

San Jose State University

Master of Science: Data Analytics

San Jose, CA

Jan. 2023 – Dec 2024

Alagappa University

Master of Business Administration: Human Resource Management

Chennai, India

Aug. 2019 – May 2021

Sathyabama University

Bachelor of Science: Computer Science Engineering

Chennai, India

Aug. 2013 – Mar 2017

TECHNICAL SKILLS

Languages: Python, SQL, C/C++, HTML/CSS, Bash/Linux, MongoDB

Cloud Platforms: AWS (S3, EC2, RDS), GCP (BigQuery, Compute Engine), Azure (Data Services)

DevOps & Tools: Jenkins, CI/CD, UDeploy, GitHub, Confluence, JIRA, Excel, Docker

Visualization & Monitoring: Tableau, Splunk, PowerBI

Libraries & Frameworks: Pandas, NumPy, PySpark, Scikit-learn, PyTorch, TensorFlow, HuggingFace Transformers, NLP, LLMs, RAG, LangChain, computer vision

EXPERIENCE

Data Scientist

May. 2024 – Current

CDUS Trading LLC

Chicago, IL

- Responsible for stock market trend analysis by leveraging technical indicators, financial reports, and historical datasets, integrating key performance indicators (KPIs) for data-driven insights.
- Reduced data preprocessing time by **30%** through **optimized ETL workflows in Apache Airflow**, accelerating model training and deployment.
- Successfully developed and deployed predictive **ML models** for daily stock market forecasting, integrating risk assessment frameworks with **NLP-based sentiment analysis**. Deployment on AWS improved accessibility and reduced latency by 50%.
- Improved decision-making speed by 40% by conducting rigorous back testing, validating model performance, and documenting insights for strategic adjustments.

Business Data Analyst

June 2017 – Nov 2020

Capgemini, Synchrony Bank (Client)

Chennai, India

- Cut incident resolution time by 35%, developing **SQL-based log analysis solutions** to proactively detect and diagnose application crashes.
- Enhanced system reliability by 40%, automating cloud service restarts and auto-scaling using **Python scripts**, minimizing manual interventions.
- Actively collaborated with cross-functional teams in Scrum calls to align on business goals, track sprint progress, and enhance operational efficiency.
- Built NLP-driven log parsers to automatically extract error patterns and critical event keywords from system logs, improving root cause analysis speed and accuracy.
- Designed and maintained **Splunk and Tableau dashboards** to monitor cloud service performance, enhancing real-time visibility into system health and supporting data-driven decision-making.

PROJECTS

• Docubot | Python, RAG, LLaMA-2, ChromaDB, openai, NLP

Developed and implemented Retrieval-Augmented Generation (RAG) using LLaMA-2 (7B), integrating ChromaDB and LangChain for optimized contextual responses. Designed an efficient pipeline leveraging quantized models and OpenAI API for enhanced retrieval and generation. [Github](#)

• Traffic Management System (TMS) | Python, ML & AI, Yolo, DeepSORT, DCRNN, ReactJS, MongoDB

Developed CCTV-based traffic flow and accident detection models using YOLOv8 for object detection. Integrated drone-based vehicle detection and tracking using YOLOv8 + DeepSORT for real-time traffic analysis. Built a multi-cluster DCRNN model for traffic speed and flow forecasting using IoT sensor data, enabling predictive analysis. [Github-Smartcity](#)

• Market Alert Bot | Python, OpenAI API, Logistic Regression, Github Actions

Developed an automated SPX market forecasting tool using Python, OpenAI, and logistic regression to analyze news sentiment, futures, and volatility data, with daily market alert emails sent via GitHub Actions.